William Chenyin

github/mutms7 | LinkedIn | Portfolio | chenyinwilliam@gmail.com | 778-903-6567

Education

University of Waterloo — Waterloo, ON
Bachelor of Computer Science (Honours), Co-operative Program — *Sep. 2025 – Apr. 2030*

- GPA: 3.9/4.0; President's Scholarship with Distinction; BC District/Authority Scholarship.
- Coursework: Algorithm Design, Object-Oriented Programming, Logic, Combinatorics, Probability.

Skills

Languages: TypeScript, JavaScript, Python, Java, C, C++, SQL, HTML/CSS
Frameworks & Libraries: React, Next.js, Node.js, FastAPI, PyTorch, Electron, Tailwind CSS
Technologies: PostgreSQL, SQLite, Supabase, Git, GitHub Actions, Gemini API, WebLLM, WebGPU

Work Experience

Software QA Automation Engineer, Co-op — May 2026 – Aug. 2026
Method Integration (Method CRM) — *Toronto, ON*

- Reduced test flakiness by 87% across three end-to-end suites by replacing fixed waits and fragile locators in **TypeScript/Playwright** tests with state-based checks and shared helpers.
- Automated 20+ iOS and Android flows in **TypeScript** using **WebdriverIO**, **Appium**, and **BrowserStack**, then added the tests to the team's daily CI pipeline.
- Turned API requirements into automated ms-scheduler-api checks for expected behavior and failure cases.
- Wrote **Playwright** and **WebdriverIO** test and review guidelines adopted by an 11-person QA team.

Projects

Typer.AI | Live App | Source | *JavaScript, Node.js, Gemini API* — Oct. 2026 – Present

- Built a **JavaScript** Manifest V3 extension for Chrome and Edge used by 25+ people across 100+ documents, suggesting words and phrases as users type, with Tab to accept and adjustable prediction confidence.
- Cut modeled AI requests by **99.4%** by caching 80 terms ranked by **Gemini 3.5 Flash-Lite** and resolving completions locally between background refreshes.
- Added a shared-key **Node.js** backend with **SQLite** request budgets; automated 19 engine and API checks and website deployment through **GitHub Actions**.

GoodLife.AI | Live App | Source | *TypeScript, React, WebLLM* — Aug. 2026 – Present

- Built a **React** and **TypeScript** habit coach used by 5+ people to plan daily actions and track habits on the web, with an **Electron** app for Windows.
- Kept AI inference on-device by integrating a quantized **Qwen2.5** 1.5B model with **WebLLM** and **WebGPU**, bundling its weights with the Windows installer for offline use.
- Added Google/email sign-in and **Supabase** sync with row-level access rules; automated Windows builds, releases, and in-app updates through **GitHub Actions**.

PaletteCard.AI | Live App | Source | *Python, PyTorch, FastAPI* — Aug. 2026 – Present

- Turned each uploaded photo into three editable greeting cards with a **Python/FastAPI** app that recognizes objects, selects matching colors, and renders designs with **Pillow**.
- Fine-tuned **MobileNetV3 Small** in **PyTorch** to recognize five object categories and trained a palette-selection neural network on 20,000 synthetic color examples.

Leadership

Vice President, The Water Boys A Cappella — Aug. 2026 – Present

- Recruited 30+ auditioners through social media promotion and in-person outreach to first-year students.

Team Captain, Waterloo Math Orientation — Jul. 2026 – Sep. 2026

- Co-led orientation for 350+ incoming students with three other captains and 40+ volunteers; gave the welcome speech and led daily activities.